

# Annotation Transfer Tool

<https://github.com/aiyshawry/Annotation-Transfer-Tool>

[Python](#)

[Computer Vision](#)

[AWS S3](#)

[CVAT](#)

[PyTorch](#)

## OVERVIEW

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A smart auto-annotation pipeline that leverages deep image similarity to transfer labels from previously human-annotated images to new unannotated ones, significantly reducing manual labeling effort in computer vision workflows.

## IMPACT

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Reduced manual annotation effort by automatically transferring labels from existing human-labeled images to new images using deep feature similarity and a modular 8-step pipeline integrated with CVAT and AWS S3.

## TECHNICAL WALKTHROUGH

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### High-level overview

This project is an Annotation Transfer Tool that significantly reduces manual labeling effort by automatically transferring annotations from previously human-labeled images to new images using deep image similarity.

In short:

**Find similar images → reuse their annotations → verify visually → export back to CVAT**

### What problem does it solve?

Manual annotation is slow and expensive. Even auto-annotation models struggle when data is limited or domain-specific. This tool leverages already annotated images and transfers their labels to new images by identifying visually similar samples using deep learning.

### Core idea

Instead of predicting bounding boxes from scratch, the tool searches for visually similar images using deep feature embeddings and transfers annotations from the closest matches.

This makes it: Faster than manual labeling More reliable than blind auto-annotation Especially effective in low-variation industrial datasets

### End-to-end pipeline explanation

#### Configuration-driven setup

The entire pipeline is controlled through a single config.yml, which defines input paths, S3 locations, and processing parameters.

This makes the pipeline: Reproducible Easy to reconfigure across datasets

#### Image & annotation ingestion

Images are downloaded from S3 based on a Datumaro JSON file exported from CVAT. Only relevant images are pulled, avoiding unnecessary data transfer.

Scripts involved: s3\_download.py datuamaru\_crop\_sort.py

#### Label normalization

Annotations are normalized into a consistent format so that downstream processes like cropping and similarity matching work reliably.

Script: normalize\_labels.py

### Crop, mask, and label generation

For each annotated object, the tool generates image crops, corresponding label files, and binary masks. This creates a reference library of labeled object appearances.

Script: `crops_generation.py`

### Deep image similarity search

A deep learning–based feature extractor generates embeddings for image crops. Using `DeepImageSearch`, the tool identifies visually similar images across the dataset.

Script: `CheckSimilarityScore_gpu.py` This is where: Feature-based search replaces brute-force labeling  
Previously labeled examples guide new annotations

### Annotation transfer & correction

Annotations from the most similar matches are transferred to the target images. Any mismatches or incorrect labels are corrected and mapped back to the original annotation files.

Script: `Correcting_matching.py` This step ensures: Label consistency Reduced noise from similarity errors

### Visualization for validation

Both normalized and denormalized annotations are visualized to verify correctness before final export.

Scripts: `visualization_normalized.py` `visualization_denormalized.py` This acts as a human-in-the-loop sanity check.

### Export back to CVAT

Finally, the updated annotations are converted back into Datumaro JSON format, making them directly uploadable to CVAT.

Script: `fetch_json_data.py`

### Two ways to run the system

- Full automation

The entire pipeline can be run with a single command (`run.py`) for batch processing.

- Step-by-step execution

Each stage can also be executed independently for debugging, validation, or partial runs.

### Interactive demo

A Jupyter notebook (`DEMO.ipynb`) demonstrates the entire workflow interactively, making it easy to understand intermediate outputs and debug results.

### One-liner summary

I built a deep similarity–based annotation transfer system that automatically reuses existing human annotations to label new images, reducing manual labeling effort while maintaining annotation quality.

### KEY DETAILS

- Downloads labeled images from AWS S3 and extracts annotations from Datumaro JSON format.

- Generates image crops, normalizes labels, and creates masks for each labeled region.

- Uses `DeepImageSearch` (GPU-accelerated) to identify the most visually similar labeled images for each new image.

- Transfers and corrects annotations from matched images, then exports back to CVAT-compatible JSON.

- Includes normalized and denormalized visualization scripts and an interactive Jupyter demo notebook.